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B. TECH
(SEM III) THEORY EXAMINATION 2017-18
SIGNALS AND SYSTEMS

Time: 3 Hours ***Total Marks: 100***

Note: **1.** Attempt all Sections. If require any missing data; then choose suitably.
2. Any special paper specific instruction.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a) Draw the signal x(t) = u(t) - 2u(t-1) + u(t-2)?
b) Define periodic and aperiodic signal with example?
c) What are the different types of system?
d) What are the dirichlet conditions?
e) State Duality property of Fourier Transform?
f) What are the drawbacks of transfer function method of representing a system?
g) Why is it not possible to construct a germanium P-N-P-N switching diode?
h) What is the relationship between Fourier transform and Laplace transform?
i) What is BIBO criterion for stability of a system?
j) Define the terms-natural response and force response of a system?

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a) Find and sketch the even and odd component of the following-
(i) tu(t) (ii) cos(ωt)u(t)

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b) Determine the z-transform and ROC of $x(n)=\left(\frac{2}{3}\right)^n u(n)+\left(-\frac{1}{2}\right)^n u(n)$

c) Find the convolution of the signals given using Fourier Transform-

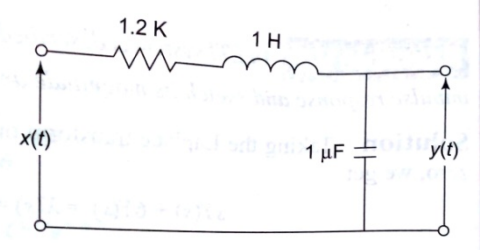
$$x_1(n)=\left(\frac{1}{2}\right)^n u(n)$$

$$x_2(n)=\left(\frac{1}{3}\right)^n u(n)$$

d) Determine the impulse response of the discrete-time system

$$y(n)-3y(n-1)+2y(n-2)=x(n)+3x(n-1)+2x(n-2)$$

e) A second order system is shown in figure.



Determine its (i) Critical resistance (ii) Damping ratio (iii) Undamped natural frequency (iv) impulse response (v) transfer function

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SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a) Sketch and check whether the given signal $x(t) = \sum_{n=-\infty}^{\infty} e^{-(2t-n)} u(2t-n)$ is periodic? If yes compute its average power?

(b) (i) Draw the signal $r(t) - 2r(t-1) + r(t-2)$?

(ii) Find signal whether the signal is energy signal, power signal or not $x(t) = e^{-3t}u(t)$.

4. Attempt any *one* part of the following: **10 x 1 = 10**

a) Prove the time shifting and Time reversal property of Z-transform?

b) (i) Find the signal whose bilateral transform is

$X(s) = \frac{1}{(s+5)(s+1)}; -5 < \mathrm{Re}(s) < -1.$

(ii) Derive the Initial Value theorem?

5. Attempt any *one* part of the following: **10 x 1 = 10**

a) Verify Parseval's theorem $\sum_{n=-\infty}^{\infty} x(n)x^*(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})X^*(e^{j\omega})d\omega$ for the following sequence $x(n) = \left(\frac{1}{2}\right)^n u(n)$

b**) (i)** What is the difference between Fourier Transform and Discrete time Fourier transform?

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(ii) Find the DTFT for the following signal $x(n) = (0.5)^n u(n) + 2^{-n}u(-n-1)$?

6. Attempt any *one* part of the following: **10 x 1 = 10**

a) Determine the stability of the system described by difference equation

$y(n) - \frac{5}{2}y(n-1) + y(n-2) = x(n) - x(n-1)$

b) (i) Define the power spectral density and their properties?

(ii) Find the cross-correlation of two finite length sequences

$x(n) = \{1,2,1,1\}; y(n) = \{1,1,2,1\}$

7. Attempt any *one* part of the following: **10 x 1 = 10**

a) For the second order system given by

$y(n) + 0.3y(n-1) - 0.1y(n-2) = x(n) - x(n-1)$ Determine (i) the system function (ii) transfer function (iii) unit sample response sequence (iv) Unit step response sequence

b) Using block diagram reduction technique, simplify

(i) The block diagram of the system given

(ii) Find the overall transfer function

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